

Multivariable Calculus

MATH 1920 Lecture Notes

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1 Coordinate Systems and Vector-Valued Functions

1.1 Coordinate Systems in 3D Space

Specifying a point or giving directions in space can be accomplished in multiple ways:

- Giving a direction and a distance.
- Using Cartesian coordinates $(d_1, d_2, d_3) = (x, y, z)$.
- Specifying an angle relative to a reference axis along with a magnitude.
- Referencing a relative landmark.
- Using geographic coordinates like latitude and longitude.

1.1.1 Polar Coordinates (2D)

In the xy -plane, a point is specified by its radial distance r from the origin and an angle θ relative to the positive x -axis: (r, θ) .

$$r = \sqrt{x^2 + y^2} \quad (r \geq 0)$$

While $\theta = \arctan\left(\frac{y}{x}\right)$ is commonly used, it is undefined when $x = 0$. The foundational algebraic transformation between systems is given by:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

1.1.2 Cylindrical Coordinates

Cylindrical coordinates (r, θ, z) extend polar coordinates into 3D space by preserving the standard Cartesian z -axis:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

1.1.3 Spherical Coordinates

Spherical coordinates are represented as (ρ, θ, ϕ) , where ρ is the distance from the origin, θ is the azimuthal angle in the xy -plane, and ϕ is the polar angle measured down from the positive z -axis.

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

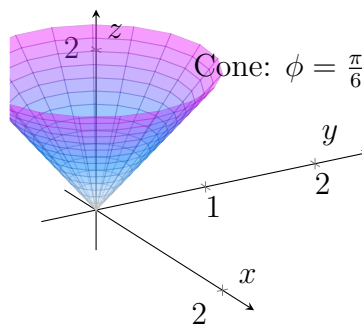
$$z = \rho \cos \phi$$

The standard domain boundaries are:

$$\rho \geq 0$$

$$0 \leq \theta < 2\pi$$

$$0 \leq \phi \leq \pi$$



1.2 Coordinate System Transformations

Problem 1.1: Conversion Between Coordinate Systems

Convert the surface equation $x^2 + \frac{y^2}{a^2} = z^2$ into cylindrical and spherical coordinates.

Solution 1.1

Cylindrical Coordinates:

Substituting $x = r \cos \theta$, $y = r \sin \theta$, and $z = z$ into the original equation yields:

$$\begin{aligned} (r \cos \theta)^2 + \frac{(r \sin \theta)^2}{a^2} &= z^2 \\ r^2 \cos^2 \theta + \frac{r^2 \sin^2 \theta}{a^2} &= z^2 \\ r^2 \left(\cos^2 \theta + \frac{\sin^2 \theta}{a^2} \right) &= z^2 \end{aligned}$$

Spherical Coordinates:

Substituting $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, and $z = \rho \cos \phi$ into the original equation yields:

$$\begin{aligned} (\rho \sin \phi \cos \theta)^2 + \frac{(\rho \sin \phi \sin \theta)^2}{a^2} &= (\rho \cos \phi)^2 \\ \rho^2 \sin^2 \phi \cos^2 \theta + \frac{\rho^2 \sin^2 \phi \sin^2 \theta}{a^2} &= \rho^2 \cos^2 \phi \\ \rho^2 \sin^2 \phi \left(\cos^2 \theta + \frac{\sin^2 \theta}{a^2} \right) &= \rho^2 \cos^2 \phi \end{aligned}$$

Assuming $\rho \neq 0$, we divide both sides by ρ^2 :

$$\begin{aligned}\sin^2 \phi \left(\cos^2 \theta + \frac{\sin^2 \theta}{a^2} \right) &= \cos^2 \phi \\ \cos^2 \theta + \frac{\sin^2 \theta}{a^2} &= \frac{\cos^2 \phi}{\sin^2 \phi} \\ \cos^2 \theta + \frac{\sin^2 \theta}{a^2} &= \cot^2 \phi\end{aligned}$$

1.3 Vector-Valued Functions

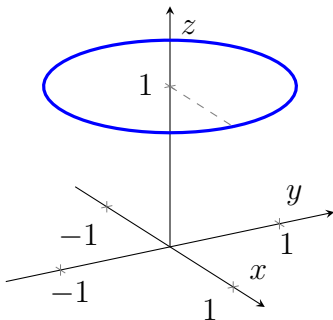
Definition 1.1: Vector-Valued Functions

A vector-valued function maps one or more real parameters to a space vector:

$$\vec{r}(t) = \langle x(t), y(t), z(t) \rangle, \quad \vec{r} : I \rightarrow \mathbb{R}^3$$

where $I \subseteq \mathbb{R}$ is an interval. This traces out a spatial curve parametrized by t .

Example 1.1. Consider the vector function $\vec{r}(t) = \langle \cos t, \sin t, 1 \rangle$ for $0 \leq t < 2\pi$. This describes a unit circle in 3D space elevated to the horizontal plane $z = 1$.



Problem 1.2: Finding Collisions and Intersections

Let $\vec{r}_1(t) = \left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle$ and $\vec{r}_2(t) = \langle t, t^2 \rangle$. Find all collision points and structural intersections between the curves.

Solution 1.2

1. Collision Points (Same location at same time t):

Setting $\vec{r}_1(t) = \vec{r}_2(t)$ gives:

$$\left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle = \langle t, t^2 \rangle$$

Equating the individual components gives the system:

$$\begin{aligned} t^2 + \frac{t}{2} &= t \\ t - t^3 &= t^2 \end{aligned}$$

Solving the first equation for t :

$$\begin{aligned} t^2 + \frac{t}{2} - t &= 0 \\ t^2 - \frac{t}{2} &= 0 \\ t \left(t - \frac{1}{2} \right) &= 0 \implies t = 0 \quad \text{or} \quad t = \frac{1}{2} \end{aligned}$$

Now, we verify both parameters in the second component equation $t - t^3 = t^2$:

$$\begin{aligned} \text{For } t = 0 : \quad 0 - (0)^3 &= (0)^2 \implies 0 = 0 \quad (\text{Valid}) \\ \text{For } t = \frac{1}{2} : \quad \frac{1}{2} - \left(\frac{1}{2} \right)^3 &= \frac{1}{2} - \frac{1}{8} = \frac{3}{8} \\ \left(\frac{1}{2} \right)^2 &= \frac{1}{4} = \frac{2}{8} \neq \frac{3}{8} \quad (\text{Invalid}) \end{aligned}$$

Therefore, $t = 0$ is the only valid collision time. Evaluating $\vec{r}_1(0)$:

$$\vec{r}_1(0) = \left\langle 0^2 + \frac{0}{2}, 0 - 0^3 \right\rangle = \langle 0, 0 \rangle$$

The collision point is $(0, 0)$.

2. Intersection Points (Same location at potentially different times t and

s):

Setting $\vec{r}_1(t) = \vec{r}_2(s)$ gives:

$$\left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle = \langle s, s^2 \rangle$$

Equating components yields:

$$s = t^2 + \frac{t}{2} \tag{1}$$

$$t - t^3 = s^2 \tag{2}$$

Substituting equation (1) into equation (2):

$$\begin{aligned} t - t^3 &= \left(t^2 + \frac{t}{2} \right)^2 \\ t - t^3 &= t^4 + t^3 + \frac{t^2}{4} \\ 0 &= t^4 + 2t^3 + \frac{t^2}{4} - t \\ 0 &= t \left(t^3 + 2t^2 + \frac{t}{4} - 1 \right) \end{aligned}$$

Setting each factor to zero:

$$t = 0 \implies s = 0^2 + \frac{0}{2} = 0$$

$\implies$ Intersection point $(0, 0)$

$$t^3 + 2t^2 + \frac{t}{4} - 1 = 0 \implies t \approx 0.576$$

Solving for s when $t \approx 0.576$:

$$s = (0.576)^2 + \frac{0.576}{2}$$

$$s \approx 0.332 + 0.288 \approx 0.620$$

Evaluating the spatial coordinates using $\vec{r}_2(s)$:

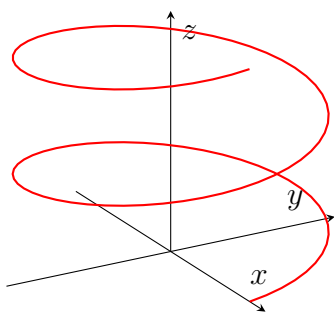
$$x = s \approx 0.620$$

$$y = s^2 \approx (0.620)^2 \approx 0.384$$

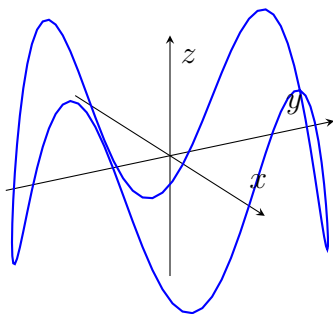
Hence, the two spatial intersection points are $(0, 0)$ and $(0.620, 0.384)$.

1.4 Gallery of Parametric Space Curves

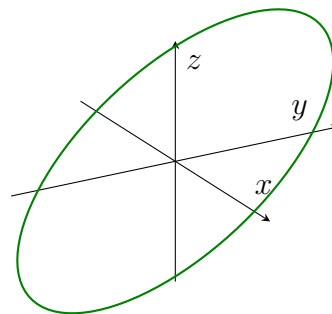
Below are standard parametric 3D space curves used in multivariable trajectory analysis:



$$\vec{r}(t) = \langle \cos t, \sin t, t \rangle$$



$$\vec{r}(s) = \langle \cos s, \sin s, \sin(4s) \rangle$$



$$\vec{r}(t) = \langle \cos t, \sin t, 4 \sin t \rangle$$